UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING DESIGN AND TECHNOLOGY

DEPARTMENT OF COMPUTING AND TECHNOLOGY

NAME:NSUBUGA EDRINE.

QUESTIONS.

a) Database anomalies are discrepancies that distort a database functionality. Base on a real life scenario example to briefly describe the three types of anomalies. Illustrations will be of added advantage

b) With two reasons and examples, briefly explain how normalization helps in mitigating redundancy and ensuring data consistency.

c) The National Identification and Registration Authority (NIRA) in Uganda maintains birth certificates that contain various data about each birth registration as shown below

BirthID	Date Of Birth/ Time	PlaceOfBirth	NAME	Sex	ParentNationality	Parents	Parent_ID_NO	District	County	Subcounty
U001	3/19/2023 2:00	Bwizi HC IV	AturindaBrighton	F	Father - Ugandan	BesigaMichael	CM7602710X3HC	Mbra	BwiziTown Council	Bwizi
					Mother - Ugandan	AkankundaJadres	CF7907405SEHN			
U002	8/15/2022 15:30	Kampala Hospital	Nabunya Sarah	F	Father - Ugandan	Nsubuga Peter	CM7601829X5HY	Kla	Kla Central	Central
					Mother - Ugandan	Namirembe Jane	CF7893058UXDY			
U003	11/21/2021 11:45	Mbarara Regional	KagwaJohn	M	Father - Ugandan	Mugerwa Paul	CM7503941X2GD	Mbra	RuharoCounty	Ruharo
					Mother - Ugandan	Nakyeyune Grace	CF7701928PXHJ			

i) Perform the normalization process from 1NF, 2NF and3NF ***(12 Marks)***

ANSWERS.

a). Take a scenario of a **University Course registration system** where each student is subjected to a specific course with each having its original Identities.
This table has as the constraints of the Attributes;

StudentID	StudentsName	CourseID	course Name	LecturerID	LecturerName

These constraints have given anomalies which may fall under each type of Anomally.

1. **Insertion Anomaly.**
   This occurs when data cannot be inserted unless other unrelated data has been added or inserted. They basically depend on a specific group of data for them to exist as one.

   Example:
   If a new Course comes up with a **CourseID, Course lecturer, Course name** … . But this Course has **no student** to study it, then the course can't exist due to the absence of the **studentID** to act as the primary key of the Database system.

2. **Deletion Anomaly.**
   This anomaly occurs after the deletion of a specific data set unintentionally and it deletes all valuable data on its delete.

   Example:
   Deletion of a student in a students' data structure might cause losing of all the data under students since the primary key is not in existence.

   This might cause loss of all data upon a given data structure.

3. **Update Anomaly.**
   This normally happens when a specific data set that appears frequently needs to be changed and must be updated multiple times.

   Example:
   If a lecturer's number needs to be changed and yet it appears more frequently. Then this means data on every row should be changed and if there is any slight act of forgetting the data change on a row, it might cause data inconsistency.
   0786594531 to 0794657839.

b). Normalization helps to clear effects of data redundancy and ensure data consistency in the following ways.

1. Minimises data redundancy.
   This is due to the data separation and grouping to avoid unnecessary reparations i.e.
   Before normalisation

Student	Course unit	lecturer	Phone number
Amon	LAN	Mr.Kasujja	0753800779
Gift	LAN	Mr.Kasujja	0753800779

   After normalisation
   A lecture table is created;

LecturerID	LecturerName	Phonenumber
D345/h34	Mr.Kasujja	0753800779

   With a primary key on it.

2. Ensures Data consistency.
   This is activated with the use of various tables which use primary keys to avoid data repetition.
   Just like a phone number in one row to avoid update repetitions.

c).Normalisation structures.

i. **<u>1NF</u>**

This way;

No repeating groups and Atomic values only

Data is grouped in tables with different parents.

Birth

(PK)Birth ID	DOB	Time	POB	ChildName	Sex	county	sub-county	District

Parent

(PK)ParentID	Parentname	ParentNumber	Nationality

Birth parent

(FK)BirthID	ParentID	ParentNumber	Parenttype

ii. **2NF**
They must be in 1NF and also no partial dependency on composite key
I.e.

BirthID and ParentID have PK.
iii. **3NF**
They must be in 2NF and also no transitive dependency

They follow existence dependency;
I.e. District to county to subcounty one existence affects the other.

iv. **4NF**
The tables are constructed.

Birth

| BirthID (PK) | DOB | Time | PlaceOfBirth | ChildName | Sex | LocationID (FK) |

Location

| LocationID (PK) | District | County | Subcounty |

Parent

| Parent_ID (PK) | ParentName | Nationality |

Birth_Parent

| BirthID (FK) | Parent_ID(FK) | ParentType |